

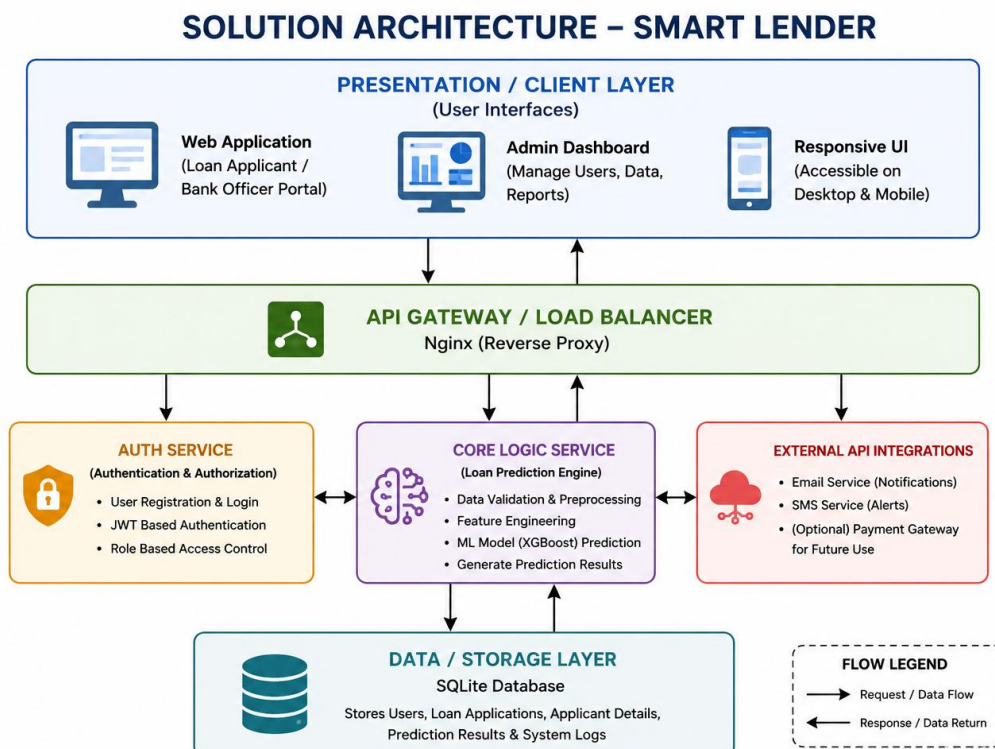
Solution Architecture

Date	11 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.

Solution Architecture Diagram :



Component Description Table

Architecture Component	Purpose in the System	Technologies / Tools
User Interface Layer	Provides a responsive interface where applicants submit loan information and administrators access prediction results and reports.	HTML5, CSS3, Bootstrap, JavaScript
Application Server	Handles client requests, validates the received information, coordinates processing, and communicates with the prediction module.	Flask (Python)
Machine Learning Engine	Cleans and transforms applicant data, extracts relevant features, executes the trained XGBoost model, and returns the loan approval prediction.	Python, Pandas, NumPy, Scikit-learn, XGBoost
Data Storage Layer	Maintains applicant records, loan application details, prediction outputs, and user-related information for future reference.	SQLite (or MySQL)
Deployment Platform	Hosts the web application, making the system accessible through a browser while supporting continuous availability and scalability.	IBM Cloud